TECHNOVA 2026 • PARTCIPANT PITCH

AKSHI-VAK

WEB-CAM EYE GAZE TYPING FOR MOTOR IMPAIRED USERS

"WHEN WORDS FAIL THE BODY, LET THE EYES SPEAK"



TEAM : NEUROQUANTA

ACCESSIBILITY AND INCLUSIVE LANGUAGE • MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY

Organized by GSA MAKAUT × WebCraftzs Technologies

01 · THE PROBLEM

A Fully Alive Mind, locked in a Frozen Body, Silenced because Eye-typing costs $5000 (~4.2L)

WHO IS AFFECTED

People with ALS / Motor Neuron Disease (MND), Spinal Cord Injury, Cerebral Palsy, Brain-Stem Stroke leading to Locked-In Syndrome, late-stage Muscular Dystrophy

All share one thing:
Eye movement is spared till the very end -
brain is sharp, body is locked

HOW BIG IT IS

India: 75,000 - 1,00,000 ALS patients alive now
Global: ~4,50,000 ALS patients, 1 dies every 90 mins
Example: A 2nd year student in Murshidabad with ALS understands coding but can't submit his project. He drops out.

WHY TODAY'S OPTIONS FALL SHORT

- Tobii Eye Tracker: Industry standard but costs $6,000; Not affordable in India, needs import.
- Bulky Hardware: Needs infrared sensor bar below monitor, not embeddable in laptops.

WHERE IT BREAKS:

- Low light in village: Tobii needs near-infrared, fails in dark room.
- With glasses/watery eyes: Commercial trackers lose reflection.
- Without Internet: Cloud tools stops

02. THE SOLUTION

AKSHI-VAK is'nt just another software app—it's a bridge - turning an ordinary laptop webcam into a voice, a shield, and an independent lifeline, proving that technology's highest purpose is to give autonomy back to those who need it most, using nothing more than the power of a glance

WHAT IT DOES

- Eye typing: gaze-dwell QWERTY keys, phrase board, local word suggestions, automatic capitalization, punctuation, delete, and clear. Learned word counts are stored locally.
- Input pages: numbers, symbols, copy/paste, full-screen, volume/brightness, Ctrl/Alt/Windows keys, and optional gaze-controlled mouse.
- Speech and accessibility: local text-to-speech (if installed), blink-to-space/speak, low-light image enhancement, screen fill-light, and optional scan mode.

WHAT MAKES IT DIFFERENT

- Our unique edge:
  Room controls (setup required): AC power/temperature, smart-light and fan actions, and a TV action.
  Caregiver and emergency tools (setup required): manual/delayed caregiver messages, emergency countdown/cancel, local siren/flash, and optional phone/SMS actions.
- Why it works better: Free, offline core: no app subscription is required; after installation, eye typing works offline on a compatible laptop/webcam. Optional device, phone/SMS, internet, or hardware services may cost extra.
- Why our team can build it: Built using lightweight Python libraries and pre-existing starter frameworks, enabling rapid prototyping without complex custom math.

03. HOW IT WORKS

FROM INPUT TO IMPACT

STEP 1: INPUT

What user gives:

- Look at key & hold 1 sec to select (dwell)
- Close eyes 0.7 - 2.5s = Insert Space, 2.5s+ = Speak text
- Turn R→Center then L→ Center = Number keypad, L→ Center then R→Center = Symbols
- 3x Right turns in 8s = Default emergency countdown, 3x Left turns in 8s = Last-dialed countdown

STEP 2: PROCESS

What system does:

- Runs eyetype_hackathon.py, camera calibration, adaptive dwell timer
- Detects: 3 blinks in 4s = emergency, 4x head shake in 2.5s = Siren-only
- Safety: 10-sec countdown + 8-sec CONFIRM CALL / CANCEL screen. Cancel = Gaze at CANCEL / press X
- ROOM setup: MENU → API SETUP → SAVE URL (use {temperature} for AC), ON/OFF URL for light/fan. Works only on ROOM page

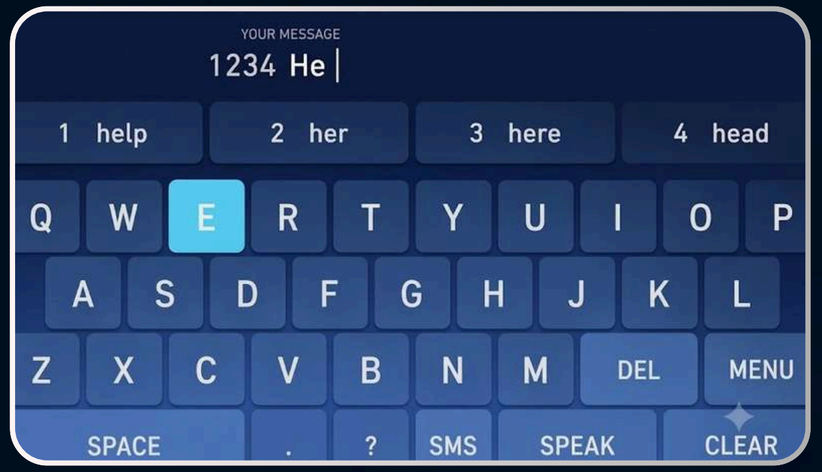
STEP 3: OUTPUT

What user gets back

- Speaks composed text, types numbers/symbols/emojis, copy/paste
- Emergency: Default call / last-dialed via ADB + Siren + flashing SOS for 10s
- Room Control: Nod up 2x = AC +1°C, Nod down 2x = AC -1°C, Tilt left 2x = Light Toggle, Tilt right 2x = Fan Toggle
- Free Offline Core: No subscription, works offline on laptop/webcam

04. DEMO

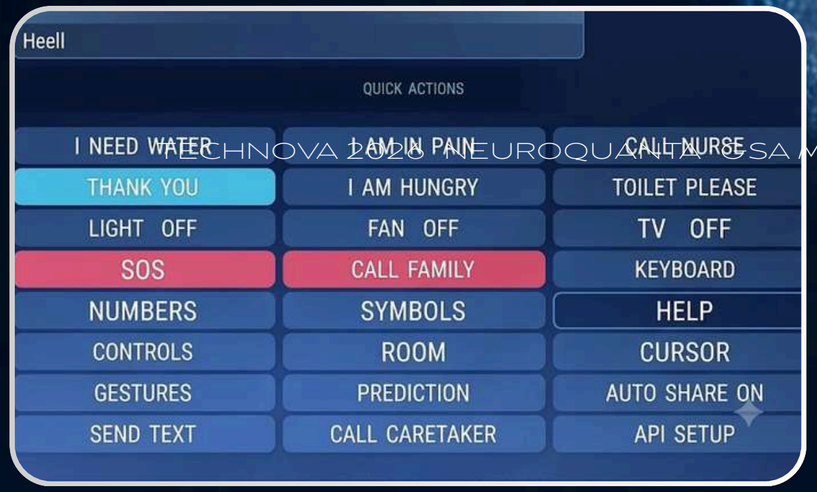
SEE IT IN ACTION



Typing numbers and letters by eye-gazing



Emergency Calls via Iris Movement



Reaching out Helpline contacts using Eye-control

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05: TECH STACK

WHAT WE BUILT IT WITH

Layer	Technology	Why We Chose it
Frontend	Pillow, pyttsx3, pyautogui, pyperclip	Pillow for UI, pyttsx3 for offline voice, pyautogui + pyperclip for typing
Backend	Python 3.12 (64-bit), opencv-contrib-python == 4.10.0.84	Main language and webcam eye-tracking
Data/AI	Mediapipe == 0.10.14, numpy<2, wordfreq	Face mesh detection + smart word prediction, offline
Tools and Hosting	pyserial, screen-brightness-control	To control AC / Light / Fan / TV via serial + control screen brightness

06. IMPACT

[90%]

Of people in low-income countries are denied access to assistive communication devices.

Who benefits and how:

Students: Build low-cost assistive tech on campus.

Campus Community: Makes campus more inclusive.

Wider City & Globally: 80M+ people with ALS / Paralysis/ Motor Neuron Disease (MND), Spinal Cord Injury, Cerebral Palsy, Brain-Stem Stroke leading to Locked-In Syndrome, late-stage Muscular Dystrophy can type, speak in their language, control appliances and SOS with just a webcam — 100% offline, free.



07. ROADMAP

WHERE WE GO NEXT



01. During the hackathon:

- Finish stable gaze-typing: 1-sec dwell, blink 0.7s space / 2.5s speak
- Test full flow: Emergency 10s + CONFIRM CALL, ROOM APIs - AC, Light, Fan, TV

02. Next 3 months

- Pilot with real users: 5-10 users at rehab center, Siliguri / Kolkata
- Fix and improve: Add Bengali + English + Hindi language support + word prediction + Android build

03. Next 12 months

- Scale to 1000+ users: Free open-source launch
- Long-term vision: Full smart-home kit under Rs.2000 + caregiver app

08. THE TEAM

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Role: Leader [Coding and App Development]

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MEET NEUROQUANTA

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SHUBHO PUJO ○ QUESTIONS?

THANK YOU

GITHUB: https://github.com/Somyajit369/AKSHI-VAK/upload/main

Live Demo: htts://www.mediafire.com/file/qz7dp86galju0f3/VID-20261011-WA0003(1).mp4/file

Guide: https://www.mediafire.com/file/wft95tix08usx7v/VID-20261011-WA0002.mp4/file

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